

1. User entity:

- Attributes:
 - **username** (Primary Key): Unique identifier for each user.
 - **name**: First (and possibly Last) Name of the user.
- Purpose: Stores information about users who sign up for the karaoke service.
- Relationship:
 - Many-to-Many relationship with karaokeFile
 - Connected to SignUp table through username.
 - The same User can sing multiple songs, meaning they can be associated with **many** Karaokefiles. On the other hand, the same Karaokefile can be sung by **many** different Users, meaning the relationship is many to many

2. KaraokeFile entity:

- Attributes:
 - **id** (Primary Key): Unique identifier for each karaoke file.
 - **version**: Version of the song associated with the karaoke file (full, duet, solo, etc....)
- Relationships:
 - Since songs can have multiple versions, each with its own file, the relationship is **many**. However, a Karaoke file can only be connected to **one** song
- Purpose: Stores information about karaoke files, including their versions and associated songs.

3. Song Entity:

- Attributes:
 - **title** (Primary Key): Title of the song.
 - **artist**: Artist or Group who performed the song.
- Relationships
 - Since songs can have multiple versions, each with its own file, the relationship is **many**. However, a karaoke file can only be connected to **one** song.
- Purpose: Stores information about songs available in the karaoke service.

4. Contributor Entity:

- Attributes:
 - **name** (Primary Key): Name of the contributor.
- Purpose: Stores names of Contributors to Songs.
- Relationship:
 - Since a Contributor may have a hand in **many** songs, as well as each Song being able to have **many** contributors, the relationship is many to many.

5. SignUp Relationship:

- Attributes:
 - **username**: Username of the user signing up.
 - **id**: Identifier of the karaoke file.
 - **qType**: type of queue, p for priority, and r for regular
 - **time**: Time of the signup.
 - **payment**: Payment amount (if any).
- Relationships:
 - Composite Primary Key (**username**, **karaokeld**): Uniquely identifies each signup entry.

- Foreign Key (**username**): References the **username** attribute in the User table.
- Foreign Key (**id**): References the **id** attribute in the KaraokeFile table.
- Purpose: Records signups made by users for songs in the karaoke service.

6. Contribution Relationship:

- Attributes:
 - **contributorName**: Name of the contributor.
 - **title**: Title of the song for which the contributor made a contribution.
 - **contrType**: Type of contribution made by the contributor (writer, bass, drummer, producer, etc....).
- Relationships:
 - Composite Primary Key (**contributorName, title**): Uniquely identifies each contribution entry.
 - Foreign Key (**contributorName**): References the **name** attribute in the Contributor table.
 - Foreign Key (**title**): References the **title** attribute in the Song table.
 - Many to Many relationship with contributor table and song. There can be **many** contributors to one song, and a single contributor can contribute to **many** different songs.
- Purpose: Stores information about contributions made by contributors to songs in the karaoke service to help with searching for songs.